Internship Project Phase 3 Report

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Project: E-Commerce Sales Performance Dashboard with Cohort Analysis

External guide:

1. **Rakshit Shetty – Director, Rooman Technologies**
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1. Introduction

This project focuses on analyzing e-commerce sales data using **Power BI** to gain meaningful insights into business performance, customer behavior, and retention trends. The objective is to transform raw transactional data into an interactive dashboard that supports data-driven decision-making.

The dashboard provides a comprehensive overview of revenue trends, product performance, customer segmentation, and retention patterns using advanced analytical techniques such as **Cohort Analysis** and **RFM Segmentation**.

2. Objectives

- Analyze overall sales performance and revenue trends
- Identify top-performing products and regions
- Segment customers based on purchasing behavior (RFM Analysis)
- Evaluate customer retention using cohort analysis
- Provide actionable insights to improve business strategy

3. Dataset Description

The dataset used in this project contains transactional data from an e-commerce platform.

Key Columns:

Column Name	**Description**
InvoiceDate	Date of transaction
Customer ID	Unique identifier for customers
Revenue	Total sales value
Quantity	Number of items purchased
Country	Customer location

Column Name	Description
Description	Product name

The dataset contains transactional records from an online retail platform covering multiple countries. It enables analysis of customer purchasing behavior and revenue trends over time.

4. Data Preprocessing

The following steps were performed to prepare the data:

- Removed missing and invalid values
- Converted date fields into proper format
- Created new calculated fields:
  - **CohortMonth** (first purchase month of customer)
  - **CohortIndex** (time difference in months)
- Aggregated data to compute retention rates
- Created measures for KPIs such as:
  - Total Revenue
  - Total Orders
  - Total Customers
  - Average Order Value

4.1 Key Measures (DAX)

- Total Revenue = SUM(Revenue)
- Total Orders = DISTINCTCOUNT(Invoice)
- Total Customers = DISTINCTCOUNT(Customer ID)
- Average Order Value = Total Revenue / Total Orders

5. Methodology

5.1 Cohort Analysis

Cohort analysis groups customers based on their first purchase month and tracks their behavior over time. This helps in understanding retention patterns and customer lifecycle.

The retention rate is calculated as the percentage of customers from the initial cohort who made repeat purchases in subsequent months.

5.2 RFM Segmentation

Customers are segmented based on:

- **Recency** – How recently a customer made a purchase
- **Frequency** – How often they purchase
- **Monetary** – How much they spend

This helps identify high-value customers and target them effectively.

6. Dashboard Overview

The dashboard consists of two main pages:

6.1 Page 1: Sales Performance Overview

This page provides a high-level summary of business performance:

- **KPI Cards:** Total Revenue, Total Orders, Total Customers, Average Order Value
- **Monthly Revenue Trend:** Shows sales growth over time
- **Top Products by Revenue:** Identifies best-performing products
- **Revenue by Country:** Highlights geographical contribution
- **Customer Segmentation (RFM):** Classifies customers into high, medium, and low value

6.2 Page 2: Customer Retention Analysis

This page focuses on customer behavior and retention:

- **Retention Decay Curve:** Shows how customer retention changes over time
- **Cohort Heatmap:** Displays retention rates across different cohorts
- **Retention KPIs:** Month 1 and Month 3 retention rates
- **Insights Panel:** Summarizes key business findings

7. Key Insights

- Revenue peaked at **£1.2M in November 2011**, indicating strong seasonal demand
- **Top 10 products contribute approximately 30% of total revenue**
- The **United Kingdom contributes around 85% of total revenue**, showing high market concentration
- Customer retention **drops significantly after 3–4 months**, indicating the need for re-engagement strategies
- Approximately **1.7K high-value customers** were identified through RFM segmentation

7.1 Hypothesis Validation

Phase 1 established four hypotheses which can now be evaluated against the findings:

- **H1 – Repeat customers contribute 60–70% of revenue:** Partially supported. The RFM analysis identified ~1,650 high-value repeat customers, and the cohort data confirms sustained repeat purchasing, though an exact revenue split was not isolated.
- **H2 – Retention decreases after first purchase:** Confirmed. The cohort heatmap shows retention drops from 100% to roughly 18–36% in Month 1 across all cohorts, with continued decline thereafter.
- **H3 – A small percentage of products drive majority revenue:** Confirmed. The top 10 products contribute approximately 30% of total revenue, consistent with a skewed sales distribution.

- **H4 – Certain regions contribute more to overall sales:** Confirmed. The United Kingdom alone accounts for ~85% of total revenue, with all other countries contributing marginally.

8. Conclusion

This project demonstrates how data analytics can drive strategic decisions in e-commerce by identifying growth opportunities, customer retention gaps, and revenue concentration risks.

The Power BI dashboard successfully transforms raw data into meaningful insights, helping identify key business opportunities. The analysis highlights the importance of customer retention, product optimization, and regional performance.

By leveraging these insights, businesses can:

- Improve customer engagement strategies
- Focus on high-value customers
- Optimize product offerings

9. Future Scope

- Implement predictive analytics for sales forecasting
- Develop customer churn prediction models
- Integrate real-time data for live dashboards
- Enhance personalization strategies using customer segmentation
- Implement recommendation systems for product suggestions

Screenshots of Dashboard

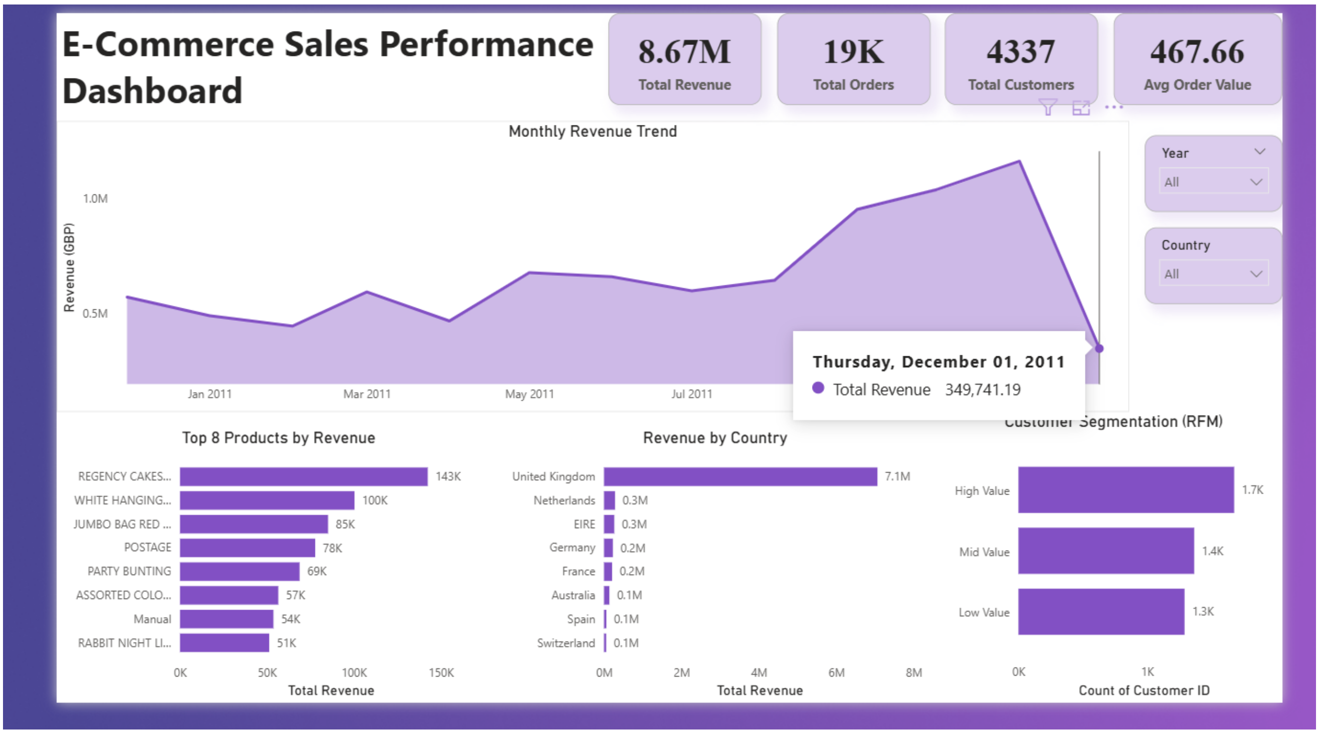

Figure 1: Sales Performance Dashboard (Page 1)

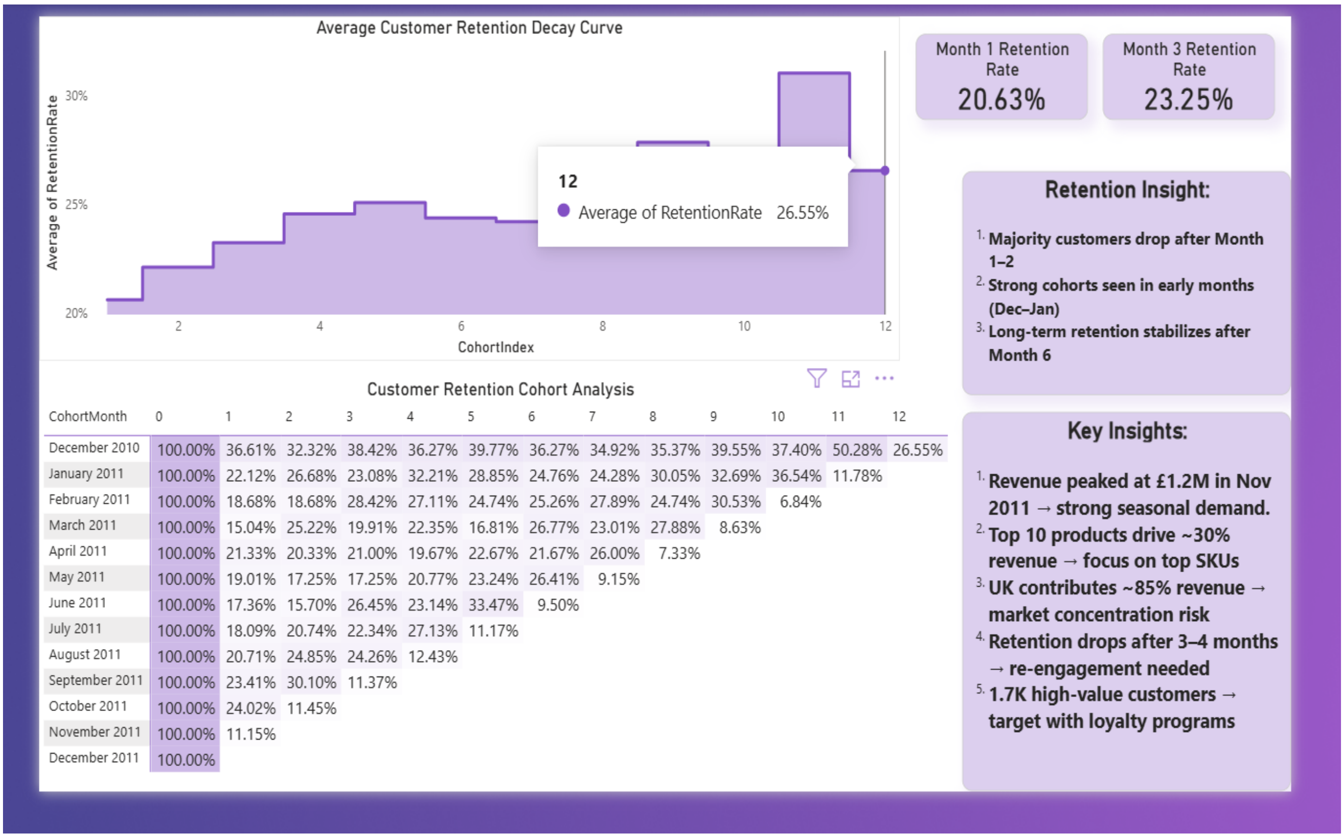

Figure 2: Customer Retention Analysis (Page 2)